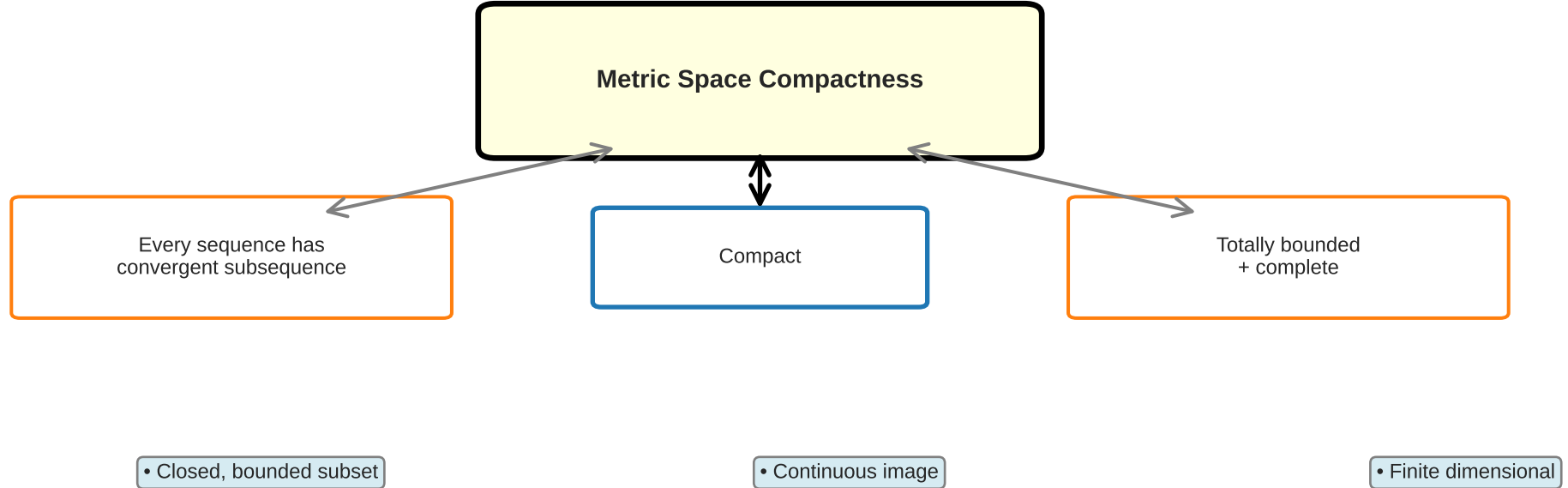


Compactness in Metric Spaces



Key Theorems:

• Heine-Borel: $\mathbb{R}^n$ is compact iff closed and bounded

• Kakutani: Unit ball in X is weakly compact iff X is reflexive